SEAT NO. CT-23025

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

SECOND YEAR (Computer Science)

SPRING SEMESTER EXAMINATIONS 2025

Time : 3 Hours | **Batch 2023** | **Dated : 03-JUN-25** | **Max Marks : 60**

Database Management Systems - CT-261

Q1. Apply normalization techniques to transform the following unnormalized table and ***create*** a set of well-structured relations up to the Third Normal Form (3NF). Discuss the anomalies addressed and resolved in the first and third normal forms. **[CLO-2][12 marks]**

Gallery Customer History Form

Customer Id: 2011

Jackson, Elizabeth
123 – 4th Avenue
Fonthill, ON
L3J 4S4

Phone (206) 284-6783

Purchases Made

Artist_ID	Artist_Name	Art_code	Art_Name	PurchaseDate	SalesPrice
03	Carol Channing	231	Laugh with Teeth	09/17/2000	7000.00
15	Dennis Frings	251	South toward Emerald Sea	05/11/2000	1800.00
03	Carol Channing	222	At the Movies	02/14/2002	5550.00
15	Dennis Frings	251	South toward Emerald Sea	07/15/2003	2200.00

The Gill Art Gallery wishes to maintain data on their customers, artists and paintings. They may have several paintings by each artist in the gallery at one time. Paintings may be bought and sold several times. In other words, the gallery may sell a painting, then buy it back at a later date and sell it to another customer.

Q2. ***Explain*** the following concepts briefly with examples. **[CLO-1][12 marks]**

1. ACID Principles.
2. Schema-less and schema-oriented database.
3. Transaction in a database and its states.
4. Deadlocking.

Q3(a). Identify the problem in the two transactions that has been created because of lack of concurrency control. ***Solve*** the following inconsistency through Two-Phase locking (2PL). **[CLO-2][6 marks]**

P.T.O

Time	Transaction T1	Transaction T2
T1	UPDATE employees SET salary = 6000 WHERE emp_id = 101;	
T2		SELECT salary FROM employees WHERE emp_id = 101; → *Reads 6000*
T3		INSERT INTO tax_log(emp_id, salary) VALUES (101, 6000);
T4		COMMIT; *(Tax log entry committed)*
T5	ROLLBACK; *(T1 aborts due to error or system crash)*	

Q3(b). ***Elaborate*** the CAP Theorem in the context of distributed database systems. For each pair of properties (Consistency, Availability, Partition Tolerance), give a real-world example of a system that prefers those two properties over the third. Justify why the third property is sacrificed in each case. **[CLO-2][6 marks]**

Q4. ***Elaborate*** the purpose of indexing in database systems by explaining how various indexing techniques are utilized in real-world applications. Provide practical examples to illustrate the implementation and benefits of each technique. **[CLO-2][12 marks]**

Q5. ***Discuss*** the primary security threats that could compromise a database system, including threats related to computer-based controls, DBMSs, and web security along with the security measures to mitigate these threats, ensuring robust database protection. **[CLO-1][12 marks]**